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TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

TMUA/C1 P1

Challenge Paper

75 minutes

PAPER 1

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open this question paper until you are told that you may do so.

A separate answer sheet is provided. Please check you have one. You also require a soft pencil and an eraser.

This booklet contains Paper 1: Applications of Mathematical Knowledge, one of the two papers in the TMUA Challenge Paper.

It has twenty multiple-choice questions and lasts 75 minutes. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

Choose the one option you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

You can use the question paper for rough working, but no extra paper is allowed. Only your responses on the answer sheet will be marked.

Dictionaries and calculators may NOT be used. Every answer is exact or a round number; if the arithmetic is turning ugly, the intended route has been missed.

Diagrams are not drawn to scale unless a question says otherwise.

Please wait to be told you may begin before turning this page.

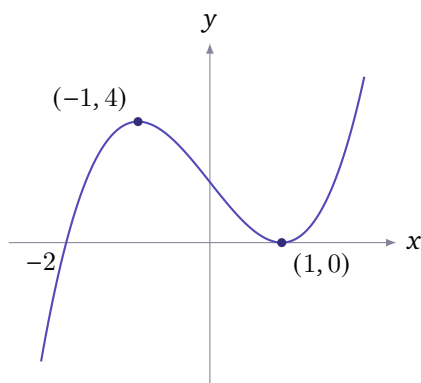
This booklet consists of twenty questions, a marked answer key, and full solutions.

PAPER 1

Applications of Mathematical Knowledge



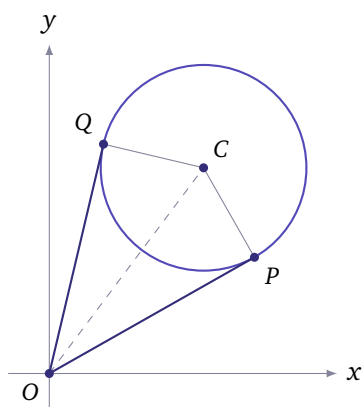
1. The diagram shows the graph of $y = f(x)$, where $f(x) = (x - 1)^2(x + 2)$.



Find the complete set of values of k for which the equation $|f(x)| = k$ has exactly four distinct real solutions.

- A) $k = 0$
- B) $0 \leq k < 4$
- C) $0 < k \leq 4$
- D) $k > 4$
- E) $0 < k < 2$
- F) $0 < k < 4$
- G) $2 < k < 4$
- H) there is no such k

2. The diagram shows the two tangents drawn from the origin O to the circle of radius 2 with centre $C = (3, 4)$, touching it at P and Q .



What is the area of the quadrilateral $OPCQ$?

- A) $\sqrt{21}$
- B) $2\sqrt{5}$
- C) $2\sqrt{21}$
- D) $4\sqrt{21}$
- E) 5
- F) 10
- G) $\frac{21}{2}$
- H) $4\sqrt{5}$

3. Let n be an integer greater than 6 for which $n - 1$ and $n + 1$ are both prime. What is the largest integer that must be a factor of $n^2(n^2 + 16)$?

- A) 12
- B) 48
- C) 120
- D) 144
- E) 240
- F) 360
- G) 720
- H) 1440

4. How many solutions does $\sin 3x = \cos 2x$ have for $0 \leq x \leq 2\pi$?

- A) 4
- B) 5
- C) 6
- D) 7
- E) 8
- F) 9
- G) 10
- H) 12

5. Evaluate $\int_0^1 x(1-x)^7 dx$.

A) $\frac{1}{72}$

B) $\frac{1}{80}$

C) $\frac{1}{64}$

D) $\frac{1}{63}$

E) $\frac{1}{56}$

F) $\frac{1}{54}$

G) $\frac{1}{9}$

H) $\frac{1}{8}$

6. How many distinct tangents to the curve $y = x^3 - 3x$ pass through the point $(0, 2)$?

- A) 0
- B) 1
- C) 2
- D) 3
- E) 4
- F) 5
- G) 6
- H) infinitely many

7. How many ordered pairs (a, b) of non-negative integers satisfy $\sqrt{a} + \sqrt{b} = \sqrt{2009}$?

- A) 0
- B) 1
- C) 2
- D) 4
- E) 7
- F) 8
- G) 14
- H) infinitely many

8. The real numbers x , y and z satisfy $x^3 + y^3 + z^3 - 3xyz = 1$. What is the smallest possible value of $x^2 + y^2 + z^2$?

A) $\frac{1}{3}$

B) $\frac{1}{2}$

C) $\frac{2}{3}$

D) 1

E) $\frac{3}{2}$

F) 2

G) 3

H) there is no smallest value

9. The line $y = mx + 1$ is a tangent to the circle $x^2 + y^2 - 6x + 8 = 0$. What is the sum of the possible values of m ?

A) $-\frac{3}{2}$

B) -1

C) $-\frac{3}{4}$

D) $-\frac{1}{2}$

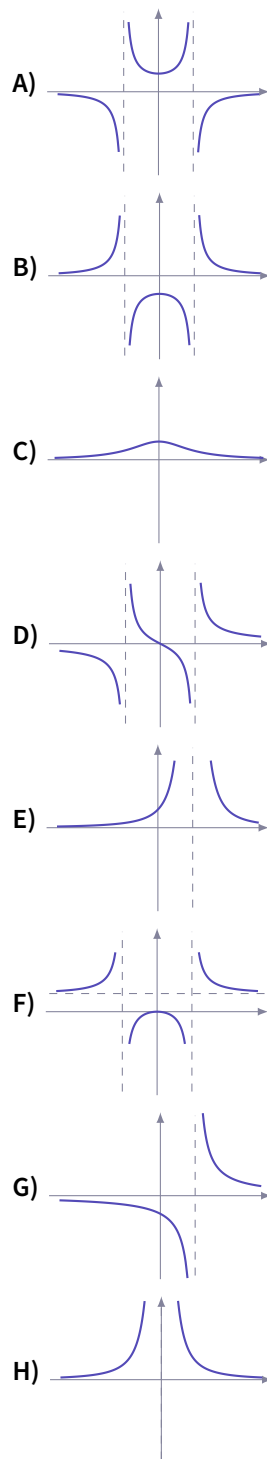
E) 0

F) $\frac{3}{4}$

G) 1

H) $\frac{3}{2}$

10. Which sketch shows the graph of $y = \frac{1}{x^2 - 1}$?



11. Find the set of real values of x for which $|x - 2| < |2x + 1|$.

A) $-3 < x < \frac{1}{3}$

B) $x < -\frac{1}{3}$ or $x > 3$

C) $\frac{1}{3} < x < 3$

D) $x < -3$

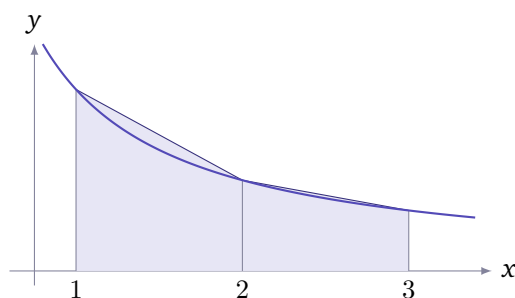
E) $x > \frac{1}{3}$

F) $-\frac{1}{3} < x < 3$

G) $x < -3$ or $x > -\frac{1}{3}$

H) $x < -3$ or $x > \frac{1}{3}$

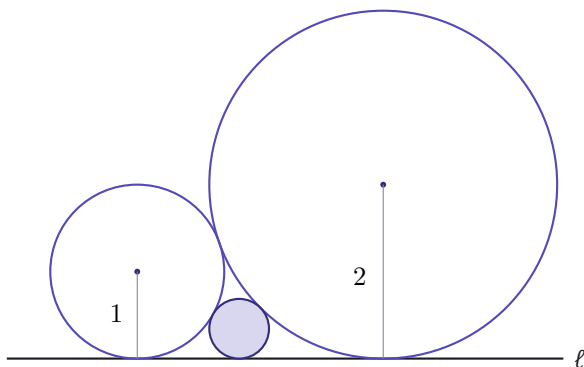
12. The trapezium rule with two strips of equal width is used to estimate $\int_1^3 \frac{1}{x} dx$, as shown.



Which statement is correct?

- A) The estimate is $\frac{7}{12}$, and it is an underestimate.
- B) The estimate is $\frac{7}{12}$, and it is an overestimate.
- C) The estimate is $\frac{4}{3}$, and it is an overestimate.
- D) The estimate is $\frac{7}{6}$, and it is an overestimate.
- E) The estimate is $\frac{7}{6}$, and it is an underestimate.
- F) The estimate is $\frac{7}{6}$, and it is exact.
- G) The estimate is $\frac{11}{6}$, and it is an overestimate.
- H) The estimate is $\frac{4}{3}$, and it is an underestimate.

- 13.** Two circles, of radii 1 and 2, touch each other and both touch the line ℓ . A third circle lies in the region between them, touching both circles and ℓ , as shown. What is its radius?



- A) $\frac{1}{3}$
- B) $\frac{2}{5}$
- C) $6 - 4\sqrt{2}$
- D) $3 - 2\sqrt{2}$
- E) $\frac{2}{9}$
- F) $\sqrt{2} - 1$
- G) $\frac{1}{2}$
- H) $\frac{2}{3}$

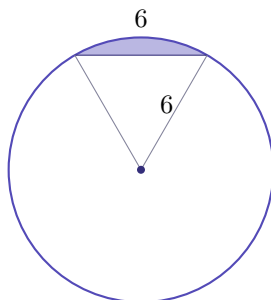
14. Find the sum of the real solutions of $4^x - 3 \cdot 2^{x+1} + 8 = 0$.

- A) 3
- B) 2
- C) 1
- D) $\log_2 3$
- E) 6
- F) 8
- G) $\log_2 6$
- H) 4

15. Find the total area of the regions enclosed between the curve $y = x^3 - x$ and the line $y = 3x$.

- A) 4
- B) $\frac{9}{2}$
- C) 6
- D) $\frac{15}{2}$
- E) 8
- F) 12
- G) 16
- H) $\frac{32}{3}$

16. A chord of length 6 is drawn in a circle of radius 6. The minor segment cut off is shaded.



What is the area of the shaded segment?

- A) $6\pi - 18$
- B) $6\pi - 9$
- C) $12\pi - 9\sqrt{3}$
- D) $9\sqrt{3} - 6\pi$
- E) $6\pi - 18\sqrt{3}$
- F) $12\pi - 18\sqrt{3}$
- G) $3\pi - 9\sqrt{3}$
- H) $6\pi - 9\sqrt{3}$

17. How many ordered pairs (m, n) of positive integers satisfy $m^2 - n^2 = 2025$?

- A) 3
- B) 4
- C) 5
- D) 6
- E) 7
- F) 8
- G) 12
- H) 15

18. An angle θ satisfies $\sin \theta + \sin^2 \theta = 1$. Find $\cos^2 \theta + \cos^4 \theta$.

A) 1

B) 0

C) $\frac{1}{2}$

D) 2

E) $\frac{3}{2}$

F) $\sqrt{5} - 2$

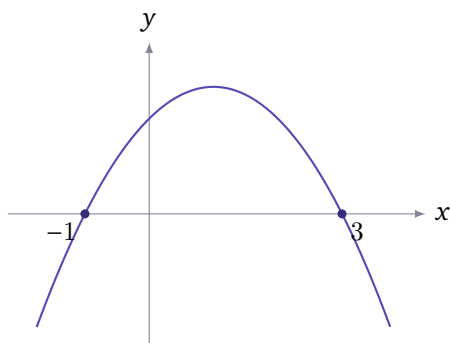
G) $\frac{3-\sqrt{5}}{2}$

H) it depends on θ

19. Find the complete set of values of k for which the equation $x|x| - 2x + k = 0$ has three distinct real roots.

- A)** $k > 0$
- B)** $0 < k < 1$
- C)** $-1 < k < 0$
- D)** $-1 < k < 1$
- E)** $0 \leq k \leq 1$
- F)** $k < 1$
- G)** $-1 \leq k \leq 1$
- H)** there is no such k

20. The diagram shows the graph of $y = ax^2 + bx + c$.



Which of the following is true?

- A) $a > 0, b > 0, c > 0$
- B) $a > 0, b > 0, c < 0$
- C) $a > 0, b < 0, c > 0$
- D) $a > 0, b < 0, c < 0$
- E) $a < 0, b < 0, c > 0$
- F) $a < 0, b < 0, c < 0$
- G) $a < 0, b > 0, c > 0$
- H) $a < 0, b > 0, c < 0$

Answer key

Paper 1			
Q	Ans	Q	Ans
1	F	11	H
2	C	12	D
3	G	13	C
4	B	14	A
5	A	15	E
6	B	16	H
7	F	17	E
8	D	18	A
9	C	19	D
10	B	20	G

Worked solutions

Paper 1: Applications of Mathematical Knowledge

1 (F). Reflect the part of the cubic below the axis to get $y = |f(x)|$, then count intersections with $y = k$. The cubic has a local maximum of 4 at $x = -1$ and a local minimum of 0 at $x = 1$, so the reflected graph has a maximum of 4 at $x = -1$, touches the axis at $x = 1$, and rises without bound on the right. Equivalently, count the roots of $f(x) = k$ and of $f(x) = -k$ separately: for $0 < k < 4$ the first gives three and the second gives one, a total of four. At $k = 4$ the line is tangent at the local maximum and the count drops to three; at $k = 0$ only the two roots of f remain; for $k > 4$ there are two. So the complete set is $0 < k < 4$. Both endpoints must be excluded, which is what separates F from B, C and G.

2 (C). The tangent length is $OP = \sqrt{OC^2 - r^2} = \sqrt{25 - 4} = \sqrt{21}$, using $OC = 5$ by Pythagoras on (3, 4). A radius is perpendicular to the tangent at the point of contact, so $OPCQ$ splits into two congruent right-angled triangles OPC and OQC , each of area $\frac{1}{2} \times \sqrt{21} \times 2 = \sqrt{21}$. The total is $2\sqrt{21}$. The kite formula $\frac{1}{2}d_1d_2$ works too, but the two-triangle split avoids finding either diagonal.

3 (G). Adapted from British Mathematical Olympiad, Round 1. Since $n - 1$ and $n + 1$ are primes greater than 5, neither is divisible by 2, 3 or 5. From the first two, n is a multiple of 6: it lies between two odd numbers so is even, and of three consecutive integers exactly one is a multiple of 3, which must be n . Write $n = 6m$. Then

$$n^2(n^2 + 16) = 36m^2(36m^2 + 16) = 144m^2(9m^2 + 4),$$

so 144 is a factor already. For the remaining factor of 5, work modulo 5: neither $n - 1$ nor $n + 1$ is a multiple of 5, so $n \equiv 0, 2$ or 3 . If $n \equiv 0$ then 5 divides n^2 ; if $n \equiv \pm 2$ then $n^2 \equiv 4$ and $n^2 + 16 \equiv 20 \equiv 0$. Either way 5 divides the product, so 720 always does. It is no better than 720: taking $n = 12$ gives $20880 = 720 \times 29$, and 29 is odd, so 1440 fails.

4 (B). Turn the cosine into a sine so that one standard family covers both cases: $\cos 2x = \sin(\frac{\pi}{2} - 2x)$. Then $\sin 3x = \sin(\frac{\pi}{2} - 2x)$ requires either $3x = \frac{\pi}{2} - 2x + 2k\pi$, giving $x = \frac{\pi}{10} + \frac{2k\pi}{5}$, or $3x = \pi - (\frac{\pi}{2} - 2x) + 2k\pi$, giving $x = \frac{\pi}{2} + 2k\pi$. In $[0, 2\pi]$ the first family supplies $\frac{\pi}{10}, \frac{\pi}{2}, \frac{9\pi}{10}, \frac{13\pi}{10}, \frac{17\pi}{10}$ and the second supplies $\frac{\pi}{2}$, which is already counted. Hence five distinct solutions. Forgetting the second family, or double counting $\frac{\pi}{2}$, is the whole trap.

5 (A). Substitute $u = 1 - x$, so $x = 1 - u$ and the limits reverse:

$$\int_0^1 x(1-x)^7 dx = \int_1^0 (1-u)u^7 du = \left[\frac{u^8}{8} - \frac{u^9}{9} \right]_1^0 = \frac{1}{8} - \frac{1}{9} = \frac{1}{72}.$$

Expanding $(1-x)^7$ term by term also works and takes ten times as long. The general result $\int_0^1 x^m(1-x)^n dx = \frac{m!n!}{(m+n+1)!}$ gives $\frac{1!7!}{9!} = \frac{1}{72}$ at sight.

6 (B). Do not start from the point; start from the point of contact. The tangent at $x = a$ has gradient $3a^2 - 3$ and passes through $(a, a^3 - 3a)$, so its equation is $y - (a^3 - 3a) = (3a^2 - 3)(x - a)$. Setting $x = 0$ gives $y = a^3 - 3a - 3a^3 + 3a = -2a^3$. For this to equal 2 we need $a^3 = -1$, so $a = -1$ and there is exactly one such tangent. The cubic $-2a^3 = 2$ has only one real root, which is what makes the count small; a point taken from inside the loop of the curve would give three.

7 (F). Adapted from British Mathematical Olympiad, Round 1. Since $2009 = 7^2 \times 41$, we have $\sqrt{2009} = 7\sqrt{41}$. If $\sqrt{a} + \sqrt{b} = 7\sqrt{41}$ then squaring gives $a + b + 2\sqrt{ab} = 2009$, so $\sqrt{ab}$ is rational and hence a and b are each 41 times a perfect square: write $a = 41s^2$ and $b = 41t^2$ with $s, t \geq 0$ integers. The equation becomes $s + t = 7$, which has eight ordered solutions, $(0, 7), (1, 6), \dots, (7, 0)$, giving

the pairs (0, 2009), (41, 1476), (164, 1025), (369, 656) and their reverses. Counting unordered pairs gives 4, and forgetting that a or b may be zero gives 6: read the question.

8 (D). Use the factorisation $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$. Put $s = x + y + z$ and $q = x^2 + y^2 + z^2$; then $xy + yz + zx = \frac{s^2 - q}{2}$, so the second bracket is $\frac{3q - s^2}{2}$ and the constraint reads $s(3q - s^2) = 2$. In particular $s > 0$ and

$$q = \frac{s^3 + 2}{3s} = \frac{s^2}{3} + \frac{2}{3s}.$$

Differentiating, $\frac{dq}{ds} = \frac{2s}{3} - \frac{2}{3s^2}$, which vanishes at $s = 1$ and changes sign from negative to positive there, so the minimum is $q = 1$. It is attained: $(x, y, z) = (1, 0, 0)$ satisfies both the constraint and $q = 1$. The factorisation is the whole problem; without it the algebra is hopeless.

9 (C). Complete the square: $x^2 + y^2 - 6x + 8 = 0$ is $(x - 3)^2 + y^2 = 1$, a circle of centre (3, 0) and radius 1. Tangency means the perpendicular distance from the centre to $mx - y + 1 = 0$ equals the radius: $\frac{|3m + 1|}{\sqrt{m^2 + 1}} = 1$, so $9m^2 + 6m + 1 = m^2 + 1$ and $8m^2 + 6m = 0$. Hence $m = 0$ or $m = -\frac{3}{4}$, and the sum is $-\frac{3}{4}$. Note that $m = 0$ is genuine: the horizontal line $y = 1$ touches the circle at its top. Discarding it as trivial gives the wrong sum.

10 (B). Read the features off the formula rather than plotting points. There are vertical asymptotes where $x^2 - 1 = 0$, at $x = \pm 1$, which rules out C, E, G and H. The function is even, which rules out D. As $|x| \rightarrow \infty$ the value tends to 0 from above, and at $x = 0$ it is $\frac{1}{-1} = -1$, so the middle branch lies below the axis and the outer branches above it: sketch B. Sketch A is $y = \frac{1}{1-x^2}$, the same curve reflected in the x -axis, and F is $y = \frac{x^2}{x^2-1}$, whose horizontal asymptote is $y = 1$ rather than $y = 0$. Parity, the asymptotes and one value at $x = 0$ separate all eight.

11 (H). Both sides are non-negative, so squaring is reversible and no cases are needed: $(x - 2)^2 < (2x + 1)^2$ gives $x^2 - 4x + 4 < 4x^2 + 4x + 1$, that is $3x^2 + 8x - 3 > 0$, or $(3x - 1)(x + 3) > 0$. A positive quadratic is positive outside its roots, so $x < -3$ or $x > \frac{1}{3}$. Squaring is legitimate precisely because of the moduli; doing it to an inequality between signed quantities is not.

12 (D). Two strips on $[1, 3]$ means $h = 1$, so the rule gives

$$\frac{h}{2} \left(f(1) + 2f(2) + f(3) \right) = \frac{1}{2} \left(1 + 2 \cdot \frac{1}{2} + \frac{1}{3} \right) = \frac{1}{2} \cdot \frac{7}{3} = \frac{7}{6}.$$

For the direction, note that $y = \frac{1}{x}$ is convex on $[1, 3]$, so every chord lies above the curve and the trapezia contain more area than the region: an overestimate. The true value is $\ln 3 \approx 1.0986$, comfortably below $\frac{7}{6} \approx 1.1667$. Convexity, not a numerical comparison, is the intended argument.

13 (C). A classic olympiad configuration. A circle of radius r touching ℓ has its centre at height r . Two such circles, of radii r_1 and r_2 , touch each other exactly when the distance between their centres is $r_1 + r_2$; by Pythagoras the horizontal separation is then

$$\sqrt{(r_1 + r_2)^2 - (r_1 - r_2)^2} = 2\sqrt{r_1 r_2}.$$

So the horizontal gap between two mutually tangent circles resting on ℓ is $2\sqrt{r_1 r_2}$, and for the small circle squeezed between the other two the gaps add: $2\sqrt{1 \cdot 2} = 2\sqrt{1 \cdot r} + 2\sqrt{2 \cdot r}$. Dividing by $2\sqrt{r}\sqrt{2}$ rearranges this to the memorable form

$$\frac{1}{\sqrt{r}} = \frac{1}{\sqrt{r_1}} + \frac{1}{\sqrt{r_2}} = 1 + \frac{1}{\sqrt{2}},$$

so $\sqrt{r} = \frac{\sqrt{2}}{\sqrt{2} + 1} = \sqrt{2}(\sqrt{2} - 1) = 2 - \sqrt{2}$ and $r = (2 - \sqrt{2})^2 = 6 - 4\sqrt{2} \approx 0.343$. Option D, $3 - 2\sqrt{2}$, is $(\sqrt{2} - 1)^2$, the answer to the same problem with two unit circles.

14 (A). Note $4^x = (2^x)^2$ and $3 \cdot 2^{x+1} = 6 \cdot 2^x$, so with $t = 2^x$ the equation is $t^2 - 6t + 8 = 0$, that is $(t - 2)(t - 4) = 0$. Both roots are positive, so both are attainable: $t = 2$ gives $x = 1$ and $t = 4$ gives $x = 2$, and the sum is 3. Adding the values of t instead of the values of x gives 6; forgetting to check positivity costs nothing here but does on the next problem of this type.

15 (E). Solve $x^3 - x = 3x$, that is $x(x^2 - 4) = 0$, so the curves meet at $x = -2, 0, 2$. The difference $g(x) = (x^3 - x) - 3x = x^3 - 4x$ is odd, so the two enclosed regions have equal area and it is enough to double one:

$$2 \int_0^2 (4x - x^3) dx = 2 \left[2x^2 - \frac{x^4}{4} \right]_0^2 = 2(8 - 4) = 8.$$

Integrating from -2 to 2 without splitting gives 0, since the signed areas cancel; that is the error the question is built around.

16 (H). Chord 6 in a circle of radius 6 makes an equilateral triangle with the two radii, so the angle at the centre is $\frac{\pi}{3}$. Using $A = \frac{1}{2}r^2(\theta - \sin \theta)$ with $r = 6$ and $\theta = \frac{\pi}{3}$,

$$A = \frac{1}{2} \times 36 \times \left(\frac{\pi}{3} - \frac{\sqrt{3}}{2} \right) = 6\pi - 9\sqrt{3} \approx 3.26.$$

Equivalently, sector minus triangle: $\frac{1}{2} \times 36 \times \frac{\pi}{3} = 6\pi$ and $\frac{\sqrt{3}}{4} \times 36 = 9\sqrt{3}$. Working in degrees and forgetting to convert is what produces the other options.

17 (E). Factorise: $(m - n)(m + n) = 2025 = 3^4 \times 5^2$. Since 2025 is odd, both factors are odd, so m and n are automatically integers whenever we split 2025 as $d \times e$ with $m = \frac{d+e}{2}$ and $n = \frac{e-d}{2}$. For n to be positive we need $d < e$, and 2025 has $(4+1)(2+1) = 15$ divisors, one of which is $\sqrt{2025} = 45$, so there are $\frac{15-1}{2} = 7$ such splittings. They give $(1013, 1012)$, $(339, 336)$, $(205, 200)$, $(117, 108)$, $(75, 60)$, $(53, 28)$ and $(51, 24)$. Counting all 15 divisors, or forgetting that $d = e = 45$ forces $n = 0$, are the two ways to miss.

18 (A). Read the condition as $\sin \theta = 1 - \sin^2 \theta = \cos^2 \theta$. Then

$$\cos^2 \theta + \cos^4 \theta = \sin \theta + \sin^2 \theta = 1.$$

No value of θ is needed, and none is available in closed form worth having: $\sin \theta = \frac{\sqrt{5}-1}{2}$. The substitution is the entire solution; solving the quadratic first and then grinding through surds gives the same answer three minutes later.

19 (D). Rewrite the equation as $g(x) = -k$, where $g(x) = x|x| - 2x$, and study g once rather than solving for each k . Splitting at the origin, $g(x) = x^2 - 2x$ for $x \geq 0$ and $g(x) = -x^2 - 2x$ for $x < 0$, so g increases on $(-\infty, -1)$, falls on $(-1, 1)$ and increases again on $(1, \infty)$, with a local maximum $g(-1) = 1$ and a local minimum $g(1) = -1$. A horizontal line meets such a curve three times exactly when it lies strictly between the two turning values, so we need $-1 < -k < 1$, that is $-1 < k < 1$. Checking $k = 0$: the roots are $-2, 0$ and 2 , three of them, as promised. Solving the two quadratics separately and forgetting to discard the roots that fall on the wrong side of the origin is what produces the one-sided options.

20 (G). Take the three signs one at a time. The parabola opens downwards, so $a < 0$. The roots are -1 and 3 , so the product of the roots is $\frac{c}{a} = -3$, giving $c = -3a > 0$; equivalently the graph crosses the y -axis above the origin, which the sketch shows. The sum of the roots is $-\frac{b}{a} = 2$, so $b = -2a > 0$. Hence $a < 0$, $b > 0$, $c > 0$. Reading b off the axis of symmetry, at $x = 1$, gives the same conclusion.

